Sec: SR.IIT_*COSC(MODEL-A&APEX) | RTA-3 | Date: 22-02-26
Time: 3 Hrs | 2022_P1 | Max. Marks: 180

KEY SHEET
MATHEMATICS

1	**2**	2	**21**	3	**2565**	4	**13**	5	**40**
6	**53**	7	**441**	8	**7**	9	**AB**	10	**CD**
11	**BC**	12	**ABC**	13	**ABD**	14	**ACD**	15	**A**
16	**B**	17	**B**	18	**A**				

PHYSICS

19	**0.25**	20	**1**	21	**168**	22	**4**	23	**2**
24	**0.60 To 0.65**	25	**2023**	26	**640**	27	**ABCD**	28	**ABC**
29	**ABD**	30	**AC**	31	**AB**	32	**AD**	33	**B**
34	**C**	35	**A**	36	**C**				

CHEMISTRY

37	**5.30 to 5.70**	38	**354**	39	**220**	40	**1.60**	41	**5.50**
42	**17**	43	**384**	44	**4**	45	**B**	46	**ABD**
47	**ABD**	48	**AB**	49	**ABCD**	50	**ABCD**	51	**A**
52	**A**	53	**D**	54	**B**				

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SOLUTIONS
MATHS

1. Let $a = 1 - x$
$b = x - y$
$c = y - z$
$d = z$

then $a + b + c + d = 1$ and $a^2 + b^2 + c^2 + d^2 = \frac{1}{4}$

$\Rightarrow (a-b)^2 + (a-c)^2 + (a-d)^2 + (b-c)^2 + (b-d)^2 + (c-d)^2 = 0$

$\Rightarrow a = b = c = d$

$\therefore\ x = \frac{3}{4},\ y = \frac{1}{2},\ z = \frac{1}{4}$

So the plane $4x + 2y + 4z + 7 = 0$ is

$\frac{3+1+1+7}{6} = 2$

2. Let $\vec{r} = x\hat{i} + y\hat{j}$

$\therefore\ \left(x\hat{i} + y\hat{j}\right) \cdot \left(-(8+x)\hat{i} + (10-y)\hat{j}\right) = 40$

$\Rightarrow -x(8+x) + y(10-y) = 40$

$\Rightarrow x^2 + y^2 + 8x - 10y + 40 = 0$

$\Rightarrow (x+4)^2 + (y-5)^2 = 1$

$\therefore\ \vec{r}$ lies on circle with centre $(-4, 5)$ and radius 1

$\left|\vec{r} + 2\hat{i} - 3\hat{j}\right|$ is distance of $\vec{r}$ from $(-2, 3)$

$\therefore$ Minimum = distance between $(-2, 3)$ and $(-4, 5)$ – radius

$= \sqrt{4+4} - 1 = 2\sqrt{2} - 1$

$\therefore\ m = \left(2\sqrt{2} - 1\right)^2 = 8 + 1 - 4\sqrt{2} = 9 - 4\sqrt{2}$

and $M = \left(2\sqrt{2} + 1\right)^2 = 9 + 4\sqrt{2}$

$\therefore\ m + M = 18$

$$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right]^2 = \begin{vmatrix} \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c} \end{vmatrix}$$

$$k = \begin{vmatrix} a^2 & \frac{ab}{2} & \frac{ac}{2} \\ \frac{ab}{2} & b^2 & \frac{bc}{2} \\ \frac{ca}{2} & \frac{cb}{2} & c^2 \end{vmatrix} = \frac{a^2b^2c^2}{2}$$

$$\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] = \frac{abc}{\sqrt{2}} \Rightarrow V = \frac{1}{6}\left[\vec{a}\ \ \vec{b}\ \ \vec{c}\right] = \frac{abc}{6\sqrt{2}}$$

Now, applying A.M. $\geq$ G.M.

$$\frac{4a^2 + 3b^2 + 2c^2}{3} \geq \left(4a^2 \cdot 3b^2 \cdot 2c^2\right)^{\frac{1}{3}}$$

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$(48)^3 \geq (4 \cdot 3 \cdot 2 \cdot 36 \cdot 2 \cdot V^2) \Rightarrow V \leq 8$

3. x_i = number of times i is used
where i = 1, 2, ..., 9
$\Rightarrow x_1 + x_2 + x_3 + ... + x_9 = 5$
$\Rightarrow$ Number of solution of this equation
$\Rightarrow {}^{13}C_5 = 1287$
Number of required numbers $= 2 \times 1287 - 9 = 2565$

4. Triangle will not be formed with (2,3,5), (2,3,6); (2,3,7), (3,4,7), (2,4,7), (2,5,7), (2,4,6)
$\therefore {}^6C_3 - 7 = 13$

5. Let $Y_0 = \{7, 14, ..., 84\} \Rightarrow$ 12 element (7λ)
$Y_1 = 1, 8, ..., 85\} \Rightarrow$ 13 element $(7\lambda + 1)$
$Y_2 = 2, 9, ..., 86\} \Rightarrow 13$ element $(7\lambda + 2)$
$Y_3 = \{3, 10, ..., 87\} \Rightarrow 13$ element $(7\lambda + 3)$
$Y_4 = \{4, 11, 18, ..., 88\} \Rightarrow 13$ element $(7\lambda + 4)$
$Y_5 = \{5, 12, ..., 89\} \Rightarrow 13$ element $(7\lambda + 5)$
$Y_6 = \{6, 13, ..., 90\} \Rightarrow 13$ element $(7\lambda + 6)$
largest Y will consist of element of Y_0 (only 1 element), Y_1, Y_2 and one of Y_3 and Y_4
$\Rightarrow 1 + 13 + 13 + 13 = 40$

6. $P(A) = \frac{3}{10}, P(B) = \frac{1}{2}, P(C) = \frac{2}{5}$

P(exactly one)= P(A) + P(B) + P(C) - 2P(A $\cap$ B) – 2P(B $\cap$ C) – 2P(C $\cap$ A) + 3P(A $\cap$ B $\cap$ C)

$$= \frac{3}{10} + \frac{1}{2} + \frac{2}{5} - 2 \times \frac{3}{10} \times \frac{1}{2} - 2 \times \frac{1}{2} \times \frac{2}{5} - 2 \times \frac{2}{3} \times \frac{3}{10} + 3 \times \frac{3}{10} \times \frac{1}{2} \times \frac{2}{5}$$

$$= \frac{5}{10} - \frac{3}{10} + 3 \times \frac{3}{10} \times \frac{1}{2} \times \frac{2}{5}$$

P (only A passing the exam) $= P(A)\, P(\bar{B})\, P(\bar{C}) = \frac{3}{10} \times \frac{1}{2} \times \frac{3}{5} = \frac{9}{100}$

$\therefore$ Required probability $= \frac{\frac{9}{100}}{\frac{44}{100}} = \frac{9}{44} = \frac{m}{n}$

7. No. of 1 steps = x, No. of 2 steps = y
$x + 2y = 7$

$$\underbrace{11.....1}_{7 \text{ times}} = \frac{7!}{7!} = 1 \; ; \; \underbrace{11..... 12}_{5 \text{ times}} = \frac{6!}{5!} = 6$$

$$11122 = \frac{5!}{3!2!} = 10$$

$$1,2,2,2 = \frac{4!}{3!} = 4$$

Total = 21 steps
Again from 9th to 16th steps
(total = 7 steps) we have
$x + 2y = 7$ (No. of ways = 21)
Total = $21 \times 21 = 441$

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8. $P\left(\frac{B}{A\cap C}\right)=\frac{P(A\cap B\cap C)}{P(A\cap C)}$

$A\cap C:(_\,\underline{T}\,_\,_\,_\,)$ or $(_\,_\,_\,\underline{T}\,_\,_)-(_\,\underline{T}\,_\,\underline{T}\,_\,_)$

$P(A\cap C)={}^5C_4\cdot\left(\frac{1}{2}\right)^5\cdot\frac{1}{2}+{}^5C_4\cdot\left(\frac{1}{2}\right)^5\cdot\frac{1}{2}-\left(\frac{1}{2}\right)^6=\frac{9}{2^6}$

$A\cap B\cap C:(\underline{H}\ \underline{T}\ \underline{H}\ \underline{H}\ \underline{H}\ \underline{T})$

or $(\underline{H}\ \underline{H}\ \underline{H}\ \underline{T}\ \underline{H}\ \underline{T})$

$P(A\cap B\cap C)=\frac{2}{2^6}$

Required probability $=\frac{\frac{2}{2^6}}{\frac{9}{2^6}}=\frac{2}{9}=\frac{m}{n}$

$\therefore n-m=7$

9. (A) Player starts with i points.

Case I: Player wins first game (probability $\frac{1}{3}$).

So, after this game player has i + 1 points and probability of winning is P_{i+1}.

Case II: Player losses first game (probability = $\frac{2}{3}$).

So, after this game player has i – 1 points and probability of winning is P_{i-1}.

Hence $P_i=\left(\frac{1}{3}\right)P_{i+1}+\left(\frac{2}{3}\right)P_{i-1}$

(B) Clearly P_0 and $P_6 = 1$. Required P_2 when n = 6.

From previous part

$P_i=\left(\frac{1}{3}\right)P_{i+1}+\left(\frac{2}{3}\right)P_{i-1}\Rightarrow P_{i+1}-P_i=2(P_i-P_{i-1})$

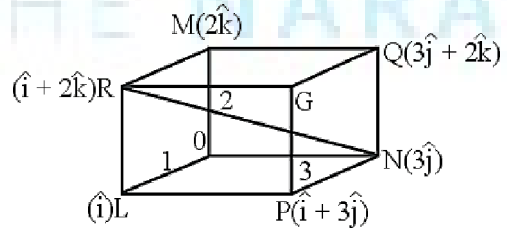
$\Rightarrow P_2-P_1=2(P_1-P_0)=2P_1\Rightarrow P_2=3P_1=(2^2-1)P_1$

$P_3-P_2=2(P_2-P_1)=4P_1\Rightarrow P_3=7P_1=(2^3-1)P_1$

Similarly, $P_4=(2^4-1)P_1$

So, $P_6=(2^6-1)P_1=1\ \Rightarrow P_1=\frac{1}{63}\ \Rightarrow P_2=3P_1=\frac{1}{21}$

10.



Equation of OL : $\vec{r}=0+\mu\hat{i}$

Equation of RN : $\vec{r}=3\hat{j}+\lambda\left(\hat{i}-3\hat{j}+2\hat{k}\right)$

(skew body diagonal)

$\Rightarrow \ell=\text{S.D}=\left|\left(3\hat{j}\right).\hat{n}\right|,\ \vec{n}=\left(\hat{i}-3\hat{j}+2\hat{k}\right)\times\hat{i}$

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$$= \left| 3\hat{j} \cdot \frac{(2\hat{j}+3\hat{k})}{\sqrt{13}} \right| = \frac{6}{\sqrt{13}}$$

Equation of ON : $\vec{r} = 0 + \mu(3\hat{j})$

Equation of QL: $\vec{r} = \hat{i} + \lambda(\hat{i} - 3\hat{j} - 2\hat{k})$

(skew body diagonal)

$\Rightarrow n = S.D = |\hat{i} . \hat{n}|, \vec{n} = (\hat{i} - 3\hat{j} - 2\hat{k}) \times \hat{j}$

$$= \left| \frac{\hat{i} \cdot (-2\hat{i} - \hat{k})}{\sqrt{5}} \right| = \frac{2}{\sqrt{5}}$$

Equation of OM: $\vec{r} = 0 + \lambda(2\hat{k})$

Equation RN: $\vec{\gamma} = 3\hat{j} + \lambda(\hat{i} - 3\hat{j} + 2\hat{k})$

(skew body diagonal)

$m = S.D. = |3\hat{j} \cdot \hat{n}|, \vec{n} = (\hat{i} - 3\hat{j} + 2\hat{k}) \times \hat{k}$

$$= \left| 3\hat{j} \frac{(-3\hat{i} - \hat{j})}{\sqrt{10}} \right| = \frac{3}{\sqrt{10}}$$

$$\Rightarrow \sqrt{13\ell^2 + 10m^2 + 5n^2} = 7$$

11. Take dot product with $\vec{a}$ we get $(\vec{r} \cdot \vec{b}) = \frac{|\vec{a}|^2}{\vec{a} \cdot \vec{b}}$

Take cross product with $\vec{b}$ we get $\frac{|\vec{a}|^2}{\vec{a}.\vec{b}}\vec{a} - (\vec{a}.\vec{b})\vec{r} = (\vec{b} \times \vec{a})$

$$\Rightarrow \vec{r} = \frac{|\vec{a}|^2}{(\vec{a}.\vec{b})^2}\vec{a} + \frac{\vec{a} \times \vec{b}}{\vec{a} . \vec{b}}$$

$$\Rightarrow f(\theta) = [\vec{r}\ \vec{a}\ \vec{b}] = \frac{|\vec{a} \times \vec{b}|^2}{\vec{a} . \vec{b}} = \frac{\sin^2\theta}{\cos\theta}$$

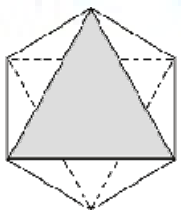
12. smallest natural number $= 2^2.3^2 = 36$

$a_1's = 1, 2, 3, 4, 6, 9, 12, 18, 36$

$\det(M) = -18$

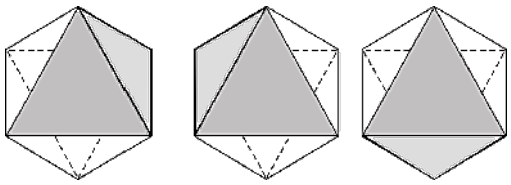
$P = 18, Q = 43, R = 112, S = 324$

13. Since the octahedron is indistinguishable by rotations, without loss of generality fix a face to be red.



There are 7! ways to arrange the remaining seven colors, but there still are three possible rotations about the fixed face, so the answer is $\frac{7!}{3} = 1680$

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14. $\frac{10!}{9!1!}\left(\frac{1}{2}\right)^1\left(\frac{1}{3}\right)^9\cdot\frac{1}{2}+\frac{10!}{8!}\left(\frac{1}{2}\right)\left(\frac{1}{3}\right)^8\left(\frac{1}{6}\right)\cdot\frac{1}{2}$

$^{10}C_1\left(\frac{1}{2}\right)^2\left(\frac{1}{3}\right)^9+{}^{10}C_2\left(\frac{1}{2}\right)^2\left(\frac{1}{3}\right)^9$

(C) $P=\frac{^{10}C_2}{^{10}C_2+{}^{10}C_1}=\frac{9}{11}$

(D) $P(W)=\frac{1}{2}P(L)=\frac{1}{6}$, $P(D)=\frac{1}{3}$

P(A wins) = (Till 10th match 1 win and 9 draw)
(11th match win) + (Till 10th match 1 win, 8 draw, 1 lost) (11th match wins)

15. (I) As there are 16 numbers between 1 and 100 which are divisible by 6. Hence required probability $\frac{^{16}C_3}{^{100}C_3}=\frac{4}{1155}$ is

(II) In the 22nd century there are 25 leap years i.e., 2100, 2104, –12195 and 75 non leap years.
Let us consider events $P(E_1)$, $P(E_2)$, P(A) order that
$P(E^1)$ - Selecting a leap year from 22nd century
$P(E^2)$ - Selecting a non-leap year from 22nd century
$P(A^2)$ - There are 53 Sundays in a year of 22nd century
Hence required probability

$$P(A)=P(E)\cdot P\left(\frac{A}{E_1}\right)+P(E_2)\cdot P\left(\frac{A}{E_2}\right)$$

(III) P(x) = 1 – no head = $1-\frac{1}{2^5}=\frac{31}{32}$

(IV) As A and $\bar{A}\cap B$ are mutually exclusive events such that

$(A\cup B)=A\cup(\bar{A}\cap B)$

$P(A\cup B)=P(A)+P(\bar{A}\cap B)$

$\frac{3}{4}=\frac{1}{3}+P(\bar{A}\cap B)=P(\bar{A}\cap B)=\frac{5}{12}$

16. P = 1 – P(all numbers are ≥ 10)

$=1-\frac{^{16}C_4}{^{25}C_4}=1-\frac{182}{1265}=\frac{1083}{1265}$

$P=\frac{^9C_1\times1\times{}^{15}C_2}{^{25}C_4}=\frac{189}{2530}$

P = P(1 even, 3 odd) + P(3 even, 1 odd)

$=\frac{^{13}C_3\ {}^{12}C_1+{}^{13}C_1\ {}^{12}C_3}{^{25}C_4}=\frac{286}{575}$

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P = 1 – P(abcd is odd)

$= 1 - \frac{^{13}C_4}{^{25}C_4} = \frac{217}{230}$

17. Total number of way of selecting 3 numbers = $^6C_3 = 20$

for A.P. $b = \frac{a+c}{2}$

$\Rightarrow$ and c are both even or both odd = $^3C_2 + {}^3C_2 = 6$ ways

for G.P. $b^2 = ac$

$\Rightarrow$ ac = 4, 9, 16, 25

ac = 4 gives a = 1, c = 4 others are not possible

for H.P. $b = \frac{2ac}{a+c}$ and b = 2, 3, 4, 5 for b = 3 we

get a = 6, c = 2

for b = 4 we get a = 6, c = 3

let a < b < c we need a + b > c

for c = 4 we get a = 2, b = 3

for c = 5 we get a = 2, b = 4 or a = 3, b = 4

for c = 6 we get a = 2, b = 5 or a = 3, b = 4 or a = 3,

b = 5 or a = 4, b = 5.

18. $(1 - x + x^2)^{30}$

$= a_0 + a_1x + a_2x^2 + \ldots\ldots\ldots + a_{60}x^{60}$ .....(1)

$x \to -x$

$(1 + x + x^2)^{30} = a_0 - a_1x + a_2x + \ldots. + a_{60}x^{60}$

$x \to \frac{1}{x}$

$(1 + x + x^2)^{30}$

$= a_0x^{60} - a_1x^{59} + \ldots\ldots. + a^{60}$ .....(2)

Multiply (1) and (2)

$(1 + x^2 + x^4)^{30}$

$= a_0 + a_1x^2 + a_2x^4 + \ldots.. + a_{60}x^{`}$ .....(3)

(I) $\left(a_0^2 - a_1^2 + a_2^2 - a_3^2 + \ldots\ldots\ldots\right)$

= coff. of x^{60} in $(1 + x^2 + x^4)^{30} = a_{30}$

$\left(\frac{a_0^2 - a_1^2 + a_2^2 - a_3^2 + \ldots..}{a_{30}}\right) = 1$

(II) Put x = 1 & –1 (1) add

$2(a_0 + a_2 + \ldots..) = 1 + 3^{30}$

$\frac{4(a_0 + a_2 + \ldots..)}{3^{30} + 1} = 2$

(III) Put x = 1 and –1 in (1) and subtract

$2(a_1 + a_3 + \ldots\ldots.) = 1 - 3^{30}$

(IV) $a_0a_1 - a_1a_2 +$

is coeff. of x^{61} in $(1 + x^2 + x^4)^{30} = 0$

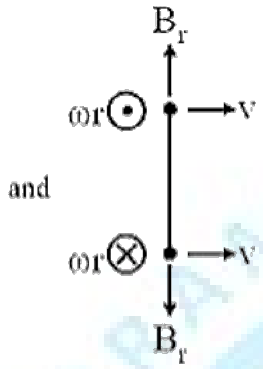
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PHYSICS

23. Due to torque created by B_r rod will acquire
Angular velocity but total energy will be conserved

$$\left(\frac{1}{2}mv_0^2\right)=\frac{1}{2}2mv^2+\frac{1}{2}\left(\frac{m\ell^2}{4}\times 2\right)\omega^2$$

$$v_0^2=v^2+\frac{\ell^2}{4}\omega^2 \qquad \ldots\ldots(i)$$



$\sum F$ due to velocity ωr will be opposite to linear velocity v

$$-q(\omega r)B_r\times 2=(2m)\left(\frac{vdv}{dx}\right)$$

From eq.(i)

$$-qB_r r\times\frac{2}{\ell}\sqrt{v_0^2-v^2}=m\frac{vdv}{dx}\left(r=\frac{\ell}{2}\right)$$

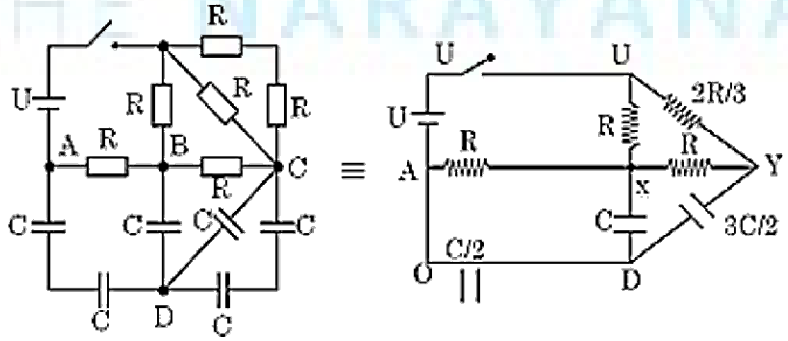
$$-qB_r\int_0^L dx=m\int_{v_0}^0\frac{vdv}{\sqrt{v_0^2-v^2}}$$

$$-\frac{qB_r}{m}L=-\int_0^{v_0}\frac{tdt}{t}\left(v_0^2-v^2=t^2\right)$$

$$\frac{qB_r}{m}L=v_0$$

$$L=\frac{mv_0}{qB_r}=2m$$

24.



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Using KCL, $\frac{x-U}{\frac{5R}{3}}+\frac{x}{R}+\frac{x-U}{R}=0$

$\Rightarrow 13x=8U \Rightarrow x=\frac{8U}{13}$

$\frac{3(Y-U)}{2R}+\frac{\left(Y-\frac{8U}{13}\right)}{R}=0 \Rightarrow \frac{5Y}{2R}=\frac{3U}{2R}+\frac{8Y}{13R}=\frac{55U}{26R}$

$\therefore\ y=\frac{11U}{13}$

$\therefore\ V_D\times\frac{C}{2}+\left(V_D-\frac{8U}{13}\right)C+\left(V_D-\frac{11U}{13}\right)\frac{3C}{2}=0$

$\frac{V_D}{2}+V_D-\frac{8U}{13}+\frac{3V_D}{2}-\frac{33U}{26}=0$

$3V_D=\frac{49U}{26}\Rightarrow V_D=\frac{49}{78}U=0.628\ U$

26. $B=\int\frac{2\mu_0.i.n.dx.R^3}{2R(R^2+x^2)^{3/2}}=$

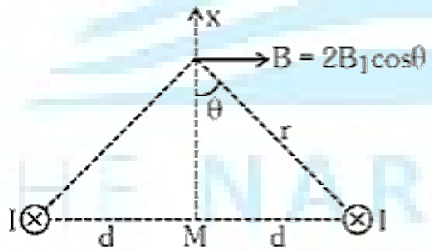
$\frac{-\mu_0 ni}{R}\int\sin^3\theta.\ R\ \cos\sec^2\theta.d\theta$

$-\mu_0 ni\int_{\theta_1}^{\theta_2}\sin\theta d\theta=\mu_0\ ni.(\cos\theta_2-\cos\theta_1)$

$4\pi\times10^{-7}\times\frac{4000}{\pi}\times2\left(\frac{4}{5}-\frac{3}{5}\right)$

$=\frac{32}{5}\times10^{-4}T=640\mu T$

29. When carrying current is same direction



$B=2\left(\frac{\mu_0 I}{2\pi\left(\frac{d}{\sin\theta}\right)}\right)\cos\theta$

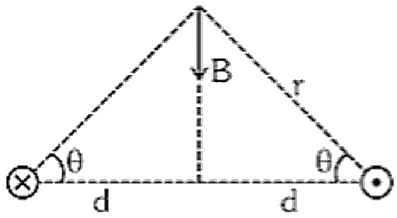
$B=\frac{\mu_0 I}{2\pi d}\sin2\theta$

$B_{min}=\frac{\mu_0 I}{2\pi d}\sin180°$ (At point M)

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$B_{max} = \frac{\mu_0 I}{2\pi d} = 1$ (When $\theta = 45°$)

Similarly, when carrying current in opposite direction

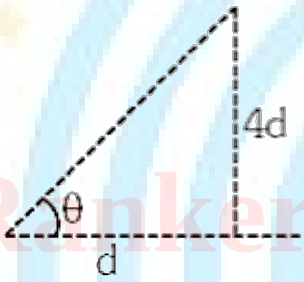


$$B = 2B_1 \cos\theta = 2\left(\frac{\frac{\mu_0 I}{2\pi d}}{\cos\theta}\right)\cos\theta$$

$$B = \frac{\mu_0 I}{\pi d}\cos^2\theta$$

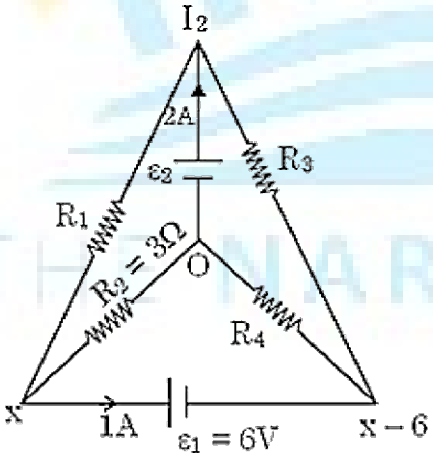
B is min when θ is max

So, min $\cos\theta = \frac{d}{\sqrt{17}d}$



$B_{min} = \frac{\mu_0 I}{\pi d}\frac{1}{17}$ and $B_{max} = \frac{\mu_0 I}{\pi d}(\cos\theta = 1)$

30. The circuit can be redrawn as



given $i_{\varepsilon_1} = 2A, i_{\varepsilon_1} = 1A$

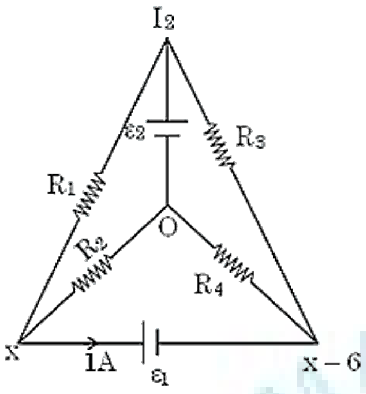
$$\therefore \frac{x}{R_2} + \frac{x-12}{R_1} + 1 = 0 \quad \text{.....(i)}$$

$$1 = \frac{x-6}{R_4} + \frac{x-6-12}{R_3} \quad \text{.....(ii)}$$

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Also, $\frac{12-x}{R_1}+\frac{12-x+6}{R_3}=2$

Also, when polarity of ε_1 is reversed,



$$\frac{12-x}{R_1}+\frac{0-x}{R_2}=1=\frac{x-6}{R_4}+\frac{x+6-12}{R_3}$$

31. For current to remain count, we have

$\frac{q}{CR}=k$ Also, at $t=0^+$ $i=\frac{q_0}{CR_0}=k$

$$\therefore \frac{qCR}{Cq_0}=R \Rightarrow R=R_0\left(1-\frac{t}{CR_0}\right)$$

$$\left[q=q_0\left(1-\frac{t}{RC_0}\right)\right]$$

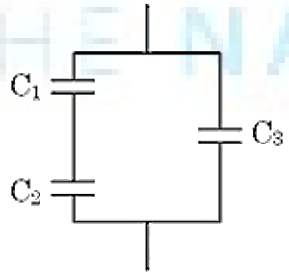
Let time at which capacitor completely discharge be t

$$\therefore \frac{q_0}{CR_0}t=q_0 \Rightarrow t=CR_0$$

Total heat developed in resistor will be equal to $\frac{q_0^2}{2C}$

32. $R_{eq}=\frac{1}{\sigma_1}\frac{d}{\frac{a}{2}\cdot a}=\frac{2d}{\sigma_1 a^2}$

The calculation of capacitance, the system can be redrawn as



$$C_1=\frac{\varepsilon_0\varepsilon_1\frac{a}{2}a}{\frac{d}{2}}$$

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$$C_2 = \frac{\varepsilon_0\varepsilon_2 \frac{a}{2}a}{\frac{d}{2}}$$

$$C_3 = \frac{\varepsilon_0\varepsilon_1 \frac{a}{2}a}{d}$$

33. Based on formula

36. $T = \frac{2\pi r^{3/2}}{\sqrt{GMe}} = 9.34\times10^3 \text{ sec}$

$$E_i = \frac{-GmM_e}{2r} - \frac{GmM_e}{2r} = \frac{-GmM_e}{r} = 20.84\times10^9 \text{ J}$$

$$E_f = \frac{-GmM_e}{r} - \frac{GmM_e}{r} = \frac{2GmM_e}{r} = 41.68\times10^9 \text{ J}$$

$L = mvr = m\sqrt{GM_e r} = 30.99\times10^{12} \text{ kg} - \text{m}^2/\text{s}$

CHEMISTRY

37. First order desorption law

$\theta = \theta_0 e^{-kt}$

$\frac{\theta_B}{\theta_A} = \frac{e^{-k_B t}}{e^{-k_A t}} = e^{(k_A - k_B)t}$

Let, $x\times10^7 = (k_A - k_B)t$

$k = Ae^{-E_a/RT}$

$R = 1.987 \text{ cal mol}^{-1} \text{ K}^{-1}$

$T = 500K$

$RT = (1.987)(500)$

$RT = 993.5 \text{ cal mol}^{-1}$

$Ea,A = 15000 \text{ cal mol}^{-1}$

$Ea,B = 25000 \text{ cal mol}^{-1}$

Surface A

$k_A = 2.0\times10^{13} e^{-15000/993.5}$

$= 2.0\times10^{13} e^{-15.10}$

$e^{-15.00} = 2.75\times10^{-7}$

$k_A = 5.50\times10^6 \text{ s}^{-1}$

Surface B

$k_B = 5.0\times10^{12} e^{-25000/993.5}$

$= 5.0\times10^{12} e^{-25.17}$

$e^{-25.17} = 5.85\times10^1$

$k_B = 5.85\times10^1 \text{ s}^{-1}$

$k_B = 58.5 \text{ s}^{-1}$

$x\times10^7 = (k_A - k_B)t$

$x\times10^7 = (5.50\times10^6 - 58.5)(10)$

Since 58.5 is negligible

$x\times10^7 = 5.50\times10^7$

$x = 5.50$

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38. Mass of KCI Solution Used

Mass = Volume x Density

$= 400 \times 1.10 = 440$ g

Mass of KCl Present

1.5% (w/w) means:

1.5 g per 100 g solution

$$\text{Mass of KCl} = \frac{1.5}{100} \times 440 = 6.60 \text{ g}$$

Moles of KCl

Molar mass of KCI = 74.5 g mol^{-1}

$$n = \frac{6.60}{74.5}$$

n = 0.0886 mol

$$\text{Coagulation value} = \frac{\text{moles of electrolyte added}}{\text{volume of sol (in L)}}$$

Volume of sol

250 mL - 0.250 L

Concentration Required for Coagulation

$$C = \frac{0.0886}{0.250}$$

C = 0.354 mol L^{-1}

0.354 × 1000 mmol L^{-1} = 354 mmol L^{-1}

Coagulation value of KCl = 354 mmol L^{-1} (ans.)

39. • Wavelength (λ): The absorption maximum $\{\lambda_{max}\}$ for $[Co(NO_2)_6]^{3-}$ is 360 nm

• Energy Calculation: The energy corresponding to this wavelength is: $E = \frac{hc}{\lambda}$

$$E = \left(\frac{6 \times 10^{-34} \times 3 \times 10^{8}}{360 \times 10^{-19}}\right) J$$

$\Delta_0 = E = 5 \times 10^{-19}$ J per molecule

$\Delta_0 = 5 \times 10^{-19} \times 6.0 \times 10^{23}$ J/mole

$\Delta_0 = 300$ KJ/mole

CFSE = $-2.4\ \Delta_0 + 2P.E = -2.4 \times 300 + 2 \times 250 = -220$ KJ/mole

40. Y = 8, X = 5

$$\frac{Y}{X} = \frac{8}{5} = \frac{16}{10} = 1.60$$

41. $2CuCl_2 + K4[Fe(CN)_6] \longrightarrow \underset{\text{Brown ppt}}{Cu\left[Fe(CN)_6\right]} + 4CKl$

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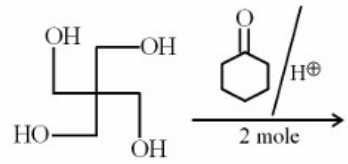
here Fe^{2+} $3d^6$ $\Rightarrow$ t_{2g}^6 e_g^0

No. unpaired electron = zero

$x = 0$

$x + 5.50 = 5.50$

42.

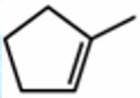


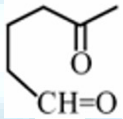
Product A



Product (C)

Number of sp^3 carbons = 17

43. Product B =  0.8 mole

Product C = 

0.8 mole

Product D = 

MF = C_6H_8O = 0.4 mole

MW of compound A = 100 ($C_6H_{12}O$) = 1 mole

MW of compound D = (C_6H_8O) = 96

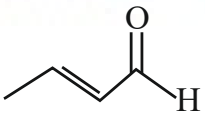
Product D in gm X = 96 × 0.4 mole

X = 38.4

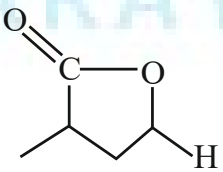
Value of 10X = 38.4 × 10 = 384

44. (i), (iii), (iv), (v)

47.



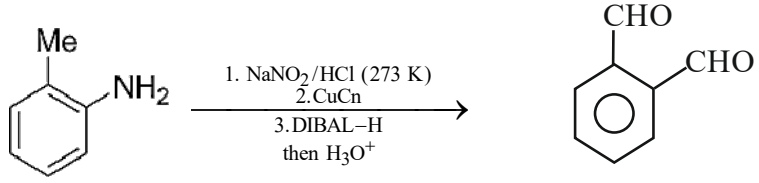
(P)



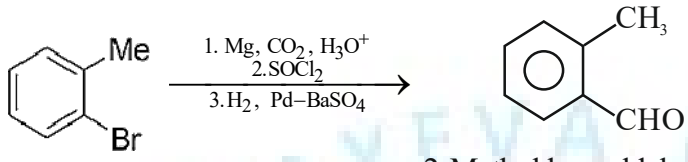
(Q)

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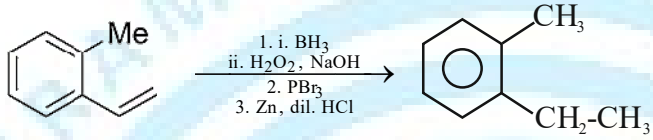
48.

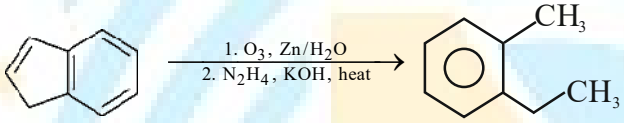
(A) 

2-Methyl benzaldehyde

(B) 

2-Methyl benzaldehyde

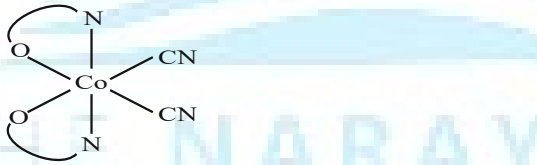
(C) 

(D) 

50. $[Cr(NH_3)_3Cl_3]$ has Facial and meridonial isomer, both are optically inactive.







Each has three combination

(i) Both CN (ii) One CN, other NC

(iii) both NC

Total 15

4 are inactive

11 are optically active

Total final isomers = 4 + 11 × 2 = 26

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